

File permissions in Linux

Project description

As a security professional it is our job to ensure authorized users have appropriate permissions so that the system can be secure. After examining the existing permissions, some users had unnecessary permission and some have not necessary permission. Using Linux command through bash shell we modified the permissions, so that appropriate users had authorization and removed any unnecessary access.

Check file and directory details

```
researcher2@e88222ccf0f0:~$ pwd
/home/researcher2
researcher2@e88222ccf0f0:~$ cd projects/
researcher2@e88222ccf0f0:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct  8 08:26 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct  8 09:15 ..
-rw--w---- 1 researcher2 research_team  46 Oct  8 08:26 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct  8 08:26 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  8 08:26 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  8 08:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_t.txt
researcher2@e88222ccf0f0:~/projects$
```

- Here changed directory to the projects directory.
- Listed all the files and directory under projects directory using "ls -la", -l displayed all the files and directories and their permission of the ownership and -a it showed all the hidden files.
- Here we have one hidden file, one subdirectory and four text files with their permissions mentioned.

Describe the permissions string

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct  8 08:26 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct  8 09:15 ..
-rw--w---- 1 researcher2 research_team  46 Oct  8 08:26 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct  8 08:26 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  8 08:26 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  8 08:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_t.txt
```

Let us take project_m.txt as an example and explain the permissions of this file using the 10 character string "-rw-r-----" which we got through the output from the command "ls -la".

- 1st character is "-", which indicates it is a file and not a directory.
- 2nd, 3rd, 4th characters are "rw-", which indicates the user has read and write permission.

- 5th, 6th, 7th characters are “r—”, which indicates the group users have only read permission.
- 8th, 9th, 10th characters are “---”, which indicates the other users have no read, write, and execute permissions for that file.

Change file permissions

```
-rw-rw-rw- 1 researcher2 research_team 46 Oct 8 04:50 project_k.txt
```

The organization does not allow others to have write access to any files. But project_k.txt has write permission for others.

```
researcher2@e88222ccf0f0:~/projects$ chmod o-w project_k.txt
researcher2@e88222ccf0f0:~/projects$ ls -l
total 20
drwx--x-- 2 researcher2 research_team 4096 Oct 8 08:26 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Oct 8 08:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_t.txt
researcher2@e88222ccf0f0:~/projects$
```

- In the first line we changed the permission of the file “project_k.txt” using “chmod o-w”. This will remove write access from the other users.
- And we confirmed it by listing it again and we can confirm that “project_k.txt” file has no write access for other users which is shown in line 5.

Change file permissions on a hidden file

```
-rw--W---- 1 researcher2 research_team 46 Oct 8 04:50 .project_x.txt
```

The research team has archived **.project_x.txt**, which is why it’s a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file.

```
researcher2@e88222ccf0f0:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@e88222ccf0f0:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 8 08:26 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 8 09:15 ..
-r--r----- 1 researcher2 research_team 46 Oct 8 08:26 .project_x.txt
drwx--x-- 2 researcher2 research_team 4096 Oct 8 08:26 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_k.txt
-rw----- 1 researcher2 research_team 46 Oct 8 08:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Oct 8 08:26 project_t.txt
researcher2@e88222ccf0f0:~/projects$
```

- In first line we changed permissions of the hidden file “.project_x.txt” using “chmod u-w,g-w,g+r”. Here “u-w” will remove write access for users, “g-w” will remove write access for group users, and “g+r” will add read permission to the group users.
- And this can be confirmed using “ls -la”, in line 5 we can see the hidden file has only read permission for both users and group.

Change directory permissions

```
drwx--x--- 2 researcher2 research_team 4096 Oct  8 04:50 drafts
```

The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents. But the group user has execute access.

```
researcher2@e88222ccf0f0:~/projects$ chmod g-x drafts
researcher2@e88222ccf0f0:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Oct  8 08:26 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_k.txt
-rw----- 1 researcher2 research_team  46 Oct  8 08:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  8 08:26 project_t.txt
researcher2@e88222ccf0f0:~/projects$
```

- In first line we used “chmod g-x”, g-x will remove execute access for the group users.
- And this can be confirmed using long listing “ls -l” and it can be seen in line 4.

Summary

After reviewing the permissions of the files and directory, we made changes so that there is no write access to any file for other users, and removed write access and added read access to the hidden file, and removed execute permission for the subdirectory to the group users. By modifying these permission we ensure authorized users have appropriate permissions so that the system can be secure.